ECON 0100 | Exercise D2 | Commons

Exercises are a certificate of the work done together in class and offers extra practice outside class.

Q1 | The Fishing Game

Two firms (Firm 1 and Firm 2) share a pond that initially contains 20,000 fish. Each firm simultaneously commits to Low Catch (L), taking one quarter of each month's starting stock, or High Catch (H), taking one half, for all three months. Both firms calculate their catches from the same starting stock. After each harvest, the surviving fish double, up to a carrying capacity of 20,000 fish. Each firm aims to maximize its total catch over three months. Following these two fishing strategies, the total catches (in thousands of fish) for the firms are summarized by the following payoff matrix.

		Firm 2	
		L	H
Firm 1	L	15, 15	8.75, 17.5
	H	17.5, 8.75	10, 10

a) What is Firm 1's best response?

b) What is Firm 2's best response?

c) Find all Nash Equilibrium.

d) Find the socially efficient strategies.

Q2 | The Fishing Game, Firm 1 Moves First

Now suppose Firm 1 commits to its three-month fishing strategy first, and Firm 2 sees Firm 1's choice before committing to its own strategy. Both firms still harvest from the same starting stock each month. Total catches are the same as in Q1. Each row of the table below is one path through the game tree: Firm 1's choice, then Firm 2's.

Firm 1 chooses	**Then Firm 2 chooses**	**Total catches (1, 2)**
L	L	15, 15
	H	8.75, 17.5
H	L	17.5, 8.75
	H	10, 10

a) If Firm 1 chooses L, what does Firm 2 choose?

b) If Firm 1 chooses H, what does Firm 2 choose?

c) Knowing how Firm 2 will respond in each case, which strategy does Firm 1 choose?

d) What is the outcome of the game, and what does each firm catch?

e) Does moving first change the outcome? Which firm has the higher payoff?